

Skills Summary

Working with Line Plots

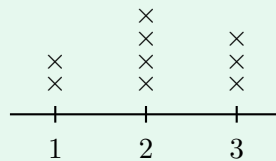
Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a Line Plot

What the parts mean: the numbers along the bottom are the *values* measured. Each × above a number is *one thing* that had that value.

Height of a column = how many things had that value. All the marks together = how many things were measured.

Example: How many pets owners in this plot have 2 pets?



Step 1. The question names one value and asks how many, so I read a single column — no adding across the plot is needed.

Step 2. Find 2 on the scale and count the marks stacked above it: there are 4.
number of owners with 2 pets = 4

Try it: A plot has 3 marks above 1, 5 marks above 2, and 2 marks above 3. How many things had the value 3?

check: 2

Skill 2 — Building a Line Plot from Data

Three steps: (1) draw a scale that covers every value in the list, with the tick marks *evenly spaced*; (2) put one × above the matching number for each value, stacking them in a column; (3) count your marks — there must be exactly as many marks as numbers in the list.

Example: Plot the list 2, 3, 3, 1.

Step 1. Every value in the list has to get its own mark, so I first make a scale wide enough for the smallest and largest values and then work through the list in order.

Step 2. The values run from 1 to 3, so the scale is 1, 2, 3. Marking each value in turn puts one × above 1, one above 2, and two above 3.

marks on the finished plot = 4, which matches the number of values in the list

Try it: Plot the list 4, 2, 4, 5, 4. How many marks should your finished plot have?

check: 5

Skill 3 — Comparing and Summarizing the Data

Most common = the tallest column.

How many more = subtract one column height from another.

Difference between greatest and least = (rightmost value with a mark) – (leftmost value with a mark). Read the *scale* for this one, not the column heights.

Example: For the list 2, 3, 3, 1, what is the difference between the greatest and the least value?

Step 1. The question asks about the values themselves, not about how often they happened, so I look along the scale rather than at the heights of the columns.

Step 2. The leftmost mark sits above 1 and the rightmost above 3, so the difference is $3 - 1$.
difference = **2**

Try it: A plot shows the values 5, 7, 7, 8, 9. What is the difference between the greatest and the least value?

check: 4

(!) Watch out: counting the *tick marks on the scale* is not the same as counting the $\times$ marks. The scale shows which values are possible; the marks show how many things were measured. Mixing them up is the most common line-plot mistake.